

Scope:

This document is to describe work instruction for Positive material identification of alloy material. Scope includes for Alloy steel material – Pipes, flanges, forgings, fittings, fasteners, gasket

1. Reference:

- Approved Inspection & Test Plan
- Approved Drawing
- Client specification - RPMS-ENGG-SPC-PP-018 Rev.0

2. Personnel Qualification:

The personnel carrying out PMI testing shall have adequate and documented theoretical and practical experience in the applied methods.

3. Equipment:

The equipment to be used to carry out PMI is of Make: THERMO SCIENTIFIC-NITON PORTABLE XRF ANALYSER.

4. Surface Condition:

The surface of the component or weld that shall be PMI tested shall be clean and reasonably smooth. Rust and / or scale as well as all kinds of oxide layers have to be removed from surface. The latter does not apply to the protective oxide layers of stainless steels.

5. Instrument Calibration:

Calibration of the instrument has been carried out by the manufacturer at their factory using various types and grades of Standard Reference Materials (SRM's,) for various types of materials. Hence no re-calibration is required to be performed.

6. Verification of Instrument:

Verification shall be performed at site once daily by using standard reference samples (it can be certified reference material or internal reference material generated using optical emission spectrometer), before commencing the actual testing or as and when required during the day.

7. Acceptance of Component & Weld:

All results within $\pm 10\%$ for base metal & for deposited weld metal shall be $\pm 12.5\%$ of the requirements of the relevant material/consumable specification shall be considered acceptable

8. Rejection of Component & Weld:

If alloy verification falls outside of the acceptance criteria using the equipment specified supplier may use chemical analysis of chips taken from the concerned material to re-verify the exact analysis of the component before final rejection. However, such procedure and the selection of the laboratory shall be subject of Client approval. The supplier shall ensure procedures for segregating rejected items and that they are properly identified to prevent reuse.

9.2. Rejection and Retest Criteria
Items or components containing materials which have not passed PMI i.e. falling outside the acceptable range shall be clearly marked as "DO NOT USE - PMI FAILED" and segregated from the remainder of the stock. Buyer shall be notified immediately for the failure.
Supplier will then have the following options with Buyer concurrence.
9.2.1. All the material, components represented by sample (same heat, lot, or identical welds) shall be scrapped or grinded out and replaced by new material, components, welds.
9.2.1.1. 100% of the remainder lot shall be tested for PMI.
9.2.1.2. 100% of the next lot of the same component shall be tested for PMI.
9.2.1.3. If next lot in above case does not identify any similar non-conformance, sampling plan shall be returned to originally specified level.
9.2.2. Correct chemistry of failed components shall be verified by laboratory chemical analysis.
9.3. Buyer/Owner shall not be responsible for cost bearing for Rejected items.

9. Documentation of Results:

A detailed report shall be prepared for the final PMI after fabrication.

10. Extent and Time of PMI Examination:

PMI Shall be performed as mentioned in Approved ITP and after final assembly before PWHT/hydro test.

Table -1

ITEMS	EXTENT OF PMI								
Heat Exchanger of Alloy steel (SS, LAS, Clad Vessel)	Material receiving: 100% PMI for alloy materials for pressure containing parts. During or after fabrication: 100% PMI for alloy materials for pressure containing parts								
Flanges, Fittings, Valves and (SS, LAS)	During or after fabrication 100% PMI for alloy materials								
Pipes (SS, LAS)	Material receiving: 100% PMI for alloy materials for pressure containing parts								
Pressure-containing welds for pressure vessels (SS, LAS, Clad Vessel)	100% PMI for alloy materials for pressure containing welds before PWHT								
Bolts/nuts (MOC SA 193 Gr. B7 & other alloy materials)	<table> <tr> <th>Lot Size</th><th>Sample Size</th></tr> <tr> <td>Upto 100</td><td>2% (Min 2)</td></tr> <tr> <td>101 to 500</td><td>1% (Min 3)</td></tr> <tr> <td>501 and above</td><td>0.5% (Min 5)</td></tr> </table>	Lot Size	Sample Size	Upto 100	2% (Min 2)	101 to 500	1% (Min 3)	501 and above	0.5% (Min 5)
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501 and above	0.5% (Min 5)								
Non-Pressure Parts (SS, LAS Internals of equipment)	100% PMI								

Minimum elements to be tested for each alloy material are given in the Table below.

Table -2

ALLOY MATERIAL	ELEMENTS
300-series austenitic stainless steels including Clad (Type 304/304L, 316/316L)	Cr, Ni, Mo (as applicable per material specification)
Low Alloy Steel (SA387 Gr 11, CL 2)	Cr, Mo.
Bolts/nuts (MOC SA 193 Gr. B7 & other alloy materials)	Cr, Mo.
100% PMI test for Metal Gaskets (RTJ), Solid Metal or jacketed metallic gaskets, inner ring & ribs of spiral wound gaskets of SS & LAS	Cr, Ni, Mo (as applicable per material specification)

11. Marking:

All alloy materials tested by PMI shall be identified using either of the following method by indicating "PMI Ok" wherever feasible.

- Bar code / Hologram sticker
- A low stress stamp marking

All grades of materials are liable for PMI test at site by Owner. In case of any discrepancies found at site, Supplier shall be liable for immediate replacement of such materials without any cost implications.

The Supplier/Contractor shall ensure that all the alloy/non-ferrous materials are segregated and stored in separately identified locations to prevent mix up of materials of different alloy specifications.
5.4. PMI examination is subject to surveillance inspection by Owner or Owner's representative